

NE523 Homework 1

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Question 1 Two-species Boltzmann Equation

The Boltzmann equation can be derived by computing the transformation that a small element in phase space sees in a short time δt . From section 2, in the absence of collisions and considering only first-order δt terms, the particles at $(\mathbf{r}, \mathbf{v})$ will stream to $\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right)$ after a short time δt , or

$$f\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right) = f(\mathbf{r}, \mathbf{v}, t_0). \quad (1)$$

Note that δt must be small enough to neglect the higher-order terms. The effect of scattering can be denoted with $\left(\frac{\partial f}{\partial t}\right)_{coll}$, the net change in f over time due to scattering (into and out of $(\mathbf{r}, \mathbf{v})$). Then in time δt the total change in f is $\left(\frac{\partial f}{\partial t}\right)_{coll} \delta t$, and

$$f\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right) = f(\mathbf{r}, \mathbf{v}, t_0) + \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t. \quad (2)$$

Expanding the left-hand side as a Taylor series in t , this becomes

$$\cancel{f(\mathbf{r}, \mathbf{v}, t)} + (\nabla_r f \cdot \mathbf{v}) \delta t + \left(\nabla_v f \cdot \frac{\mathbf{F}}{m}\right) \delta t + \left(\frac{\partial f}{\partial t}\right) \delta t = \cancel{f(\mathbf{r}, \mathbf{v}, t)} + \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t \quad (3)$$

$$\left(\mathbf{v}\nabla_r + \frac{\mathbf{F}}{m}\nabla_v + \frac{\partial}{\partial t}\right) f(\mathbf{r}, \mathbf{v}, t) \delta t = \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t \quad (4)$$

$$\left(\mathbf{v}\nabla_r + \frac{\mathbf{F}}{m}\nabla_v + \frac{\partial}{\partial t}\right) f(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial f}{\partial t}\right)_{coll}. \quad (5)$$

The same applies for particles of species g , so

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{F}}{m} \nabla_v + \frac{\partial}{\partial t} \right) f(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial f}{\partial t} \right)_{coll} \quad (6)$$

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{G}}{m} \nabla_v + \frac{\partial}{\partial t} \right) g(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial g}{\partial t} \right)_{coll}. \quad (7)$$

The collision integral can be found by computing the rate of scattering into or out of $(\mathbf{r}, \mathbf{v})$ due to collisions of particles of both species.

Question 2 Differential Volume Invariant

Question 3 Maxwell-Boltzmann Moments